UCCN 2003

UCCN 2243

TCP/IP Internetworking, Internetwork Principles & Practices

(Lecture 01)

Review & Upgrade of IPv4 Subnets Rules

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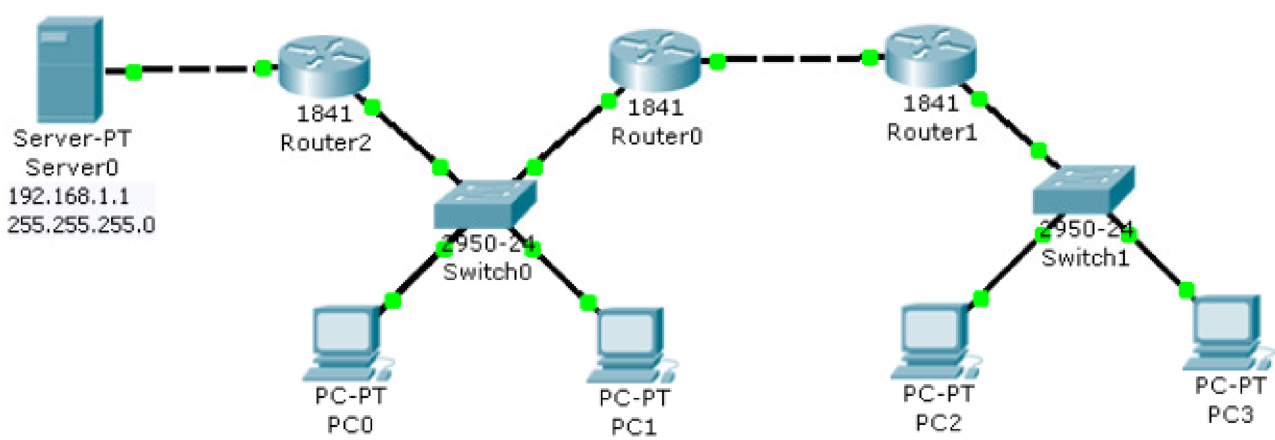
Review of IP Subnet & LAN

IP subnets = LAN?

UCCN1004: IP Subnet & LAN

- In UCCN1004, we treat LAN = IP subnet
  - Multiple LANs forms an enterprise network or campus network
- LAN is the “hardware” part of the network, “formed” with
  - End devices connected to a switch/hub/access point,
  - Bound by routers
- IP Subnet is the “software side” of LAN
  - <u>There are rules in setting IP addresses in LAN so that the network elements can communicate with each other.</u>



IP Subnet = Layer-3 ; LAN = Layer-2

- IP subnet is a layer-3 mechanism,
  - IP addresses has to be configured.
- LAN is a layer-2 mechanism,
  - It involves “domain” mechanism such as collision or broadcast.
- Beneath this layer-3 of IP subnets, there are both layer-2 and layer-1 technologies that are supporting it.
  - Layer-2 technology
    - Frame Relay, Point-to-Point Protocol (PPP), Ethernet, PPPoE, PPPoA, etc
  - Layer-1 technology
    - Telephone network, Cable TV network, Satellite network, Fiber Network, etc

UCCN1004 to UCCN2243: IP Subnets

- We need to upgrade the concept IP subnet that includes WAN (wide area network)
  - IP subnet = LAN and WAN
- In LAN (local area network), IP subnets is typically formed with:
  - a hub
  - an Ethernet switch,
  - a wireless access point ,
  - point to point network .
- In WAN (wide area network – network comprising a larger boundary), IP subnet is formed with
  - a layer-2 technology switch or network (e.g. frame relay)
  - a point-to-point phone line with modem (or layer 1 device)
  - Or the combination of the both (layer 1 + layer 2)
- In general, IP subnet comprised of:
  - IP interfaces connected to a layer-2 devices (either WAN or LAN), or point-to-point connections.
  - All IP addresses of the interfaces share a same network ID.

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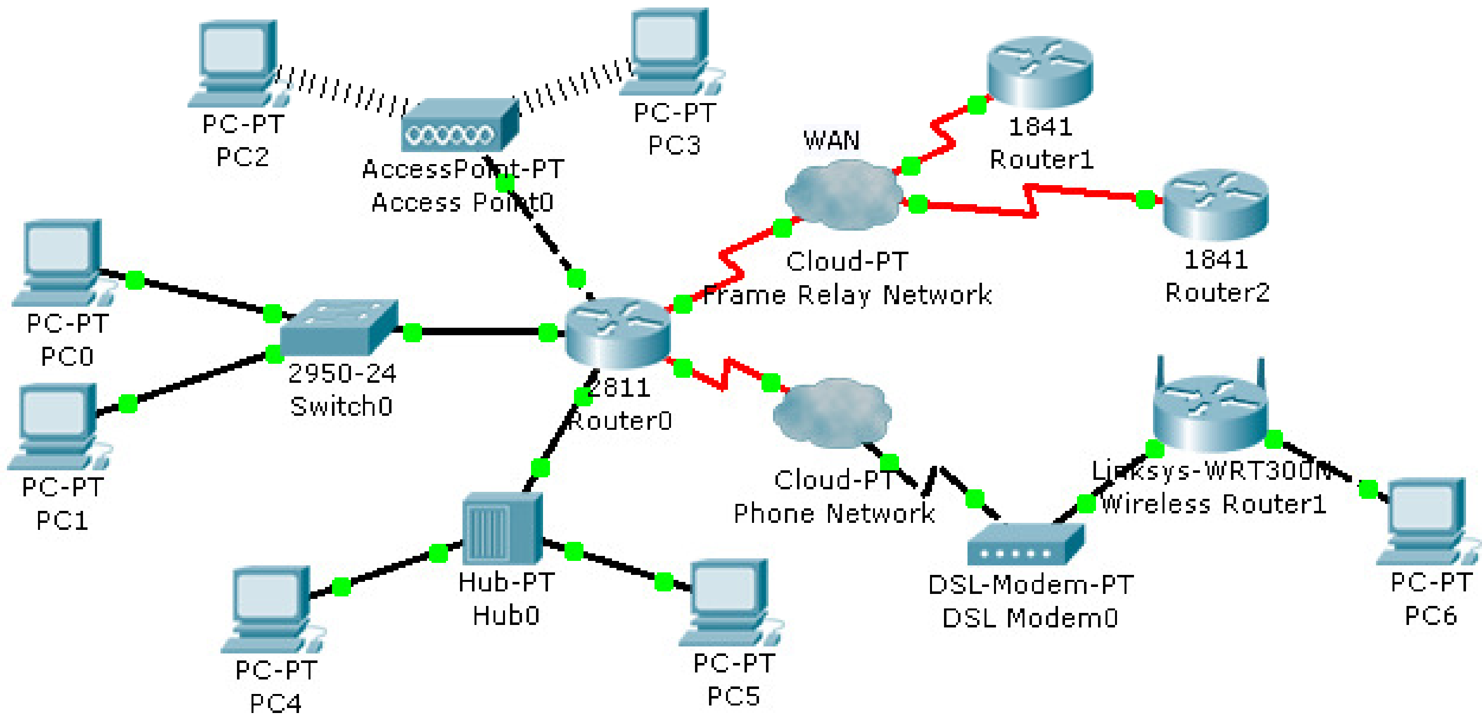
Quick Quiz: Visualization of IP Subnet

- How many subnets are there in this network?
- Could you give a brief description of the nature of these subnets? (e.g. LAN, WAN)

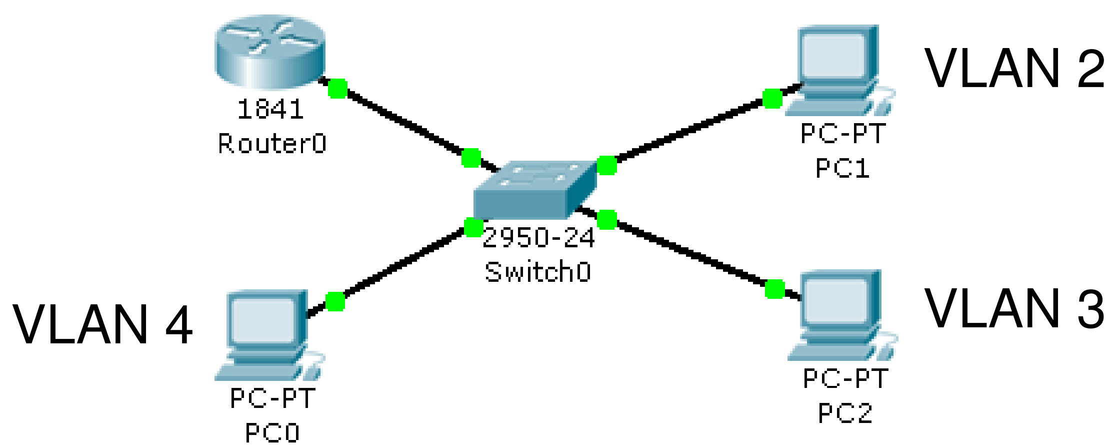


Answer

- 6 subnets
- PC0 & PC1, LAN formed by Ethernet switch
- PC2 & PC3, wireless LAN formed by a wireless access point.
- PC4 & PC5, LAN formed by a Hub
- Router0 & Router1 & Router2, WAN formed by a frame relay network
- Wireless Router1 & Router0, WAN, point-to-point, with modem through a phone network and
- PC6 to Wireless Router1, LAN, point-to-point network.

UCCN1004: IP Subnet & VLAN

- VLAN does not violate the definition of IP subnet.
- How many IP addresses do you require in the following network, with each PC belongs to a different VLAN (meaning there is 3 VLAN in this network)?
- How many network IDs are there in this network?



Answer

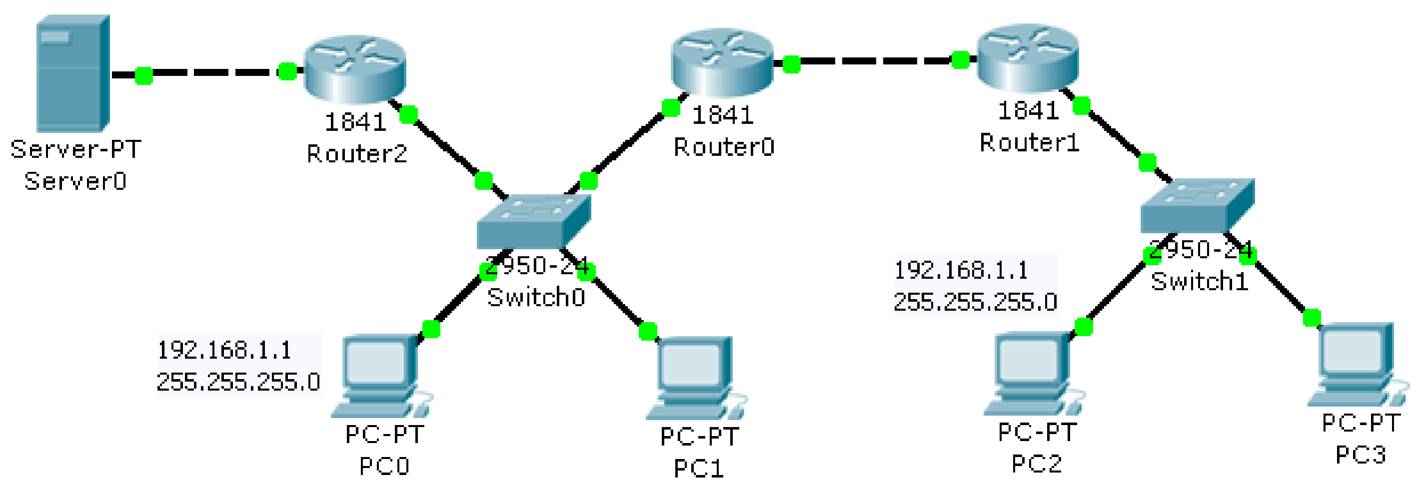
- 6 IP addresses
  - The Router0 to switch0 connection is in trunk mode.
  - It needs 3 sub-interfaces, though 1 cable, hence 3 gateway IP
  - Each PC requires 3 IPs
  - 3 gateway IP + 3 PC IPs = 6 IP
- 3 network IDs since there are 3 VLAN

IP Subnet Rule #1

Do IP addresses have to be unique within a closed network?

UCCN1004: IP Subnet Rule #1

- Every IP address within the "closed" network has to be unique.
  - There can't be two same IP addresses in the network
  - Same applies to the Public IP addresses in the Internet



PC0 and PC2 can't have the same IP in this network

UCCN2243: IP Subnet Rule #1

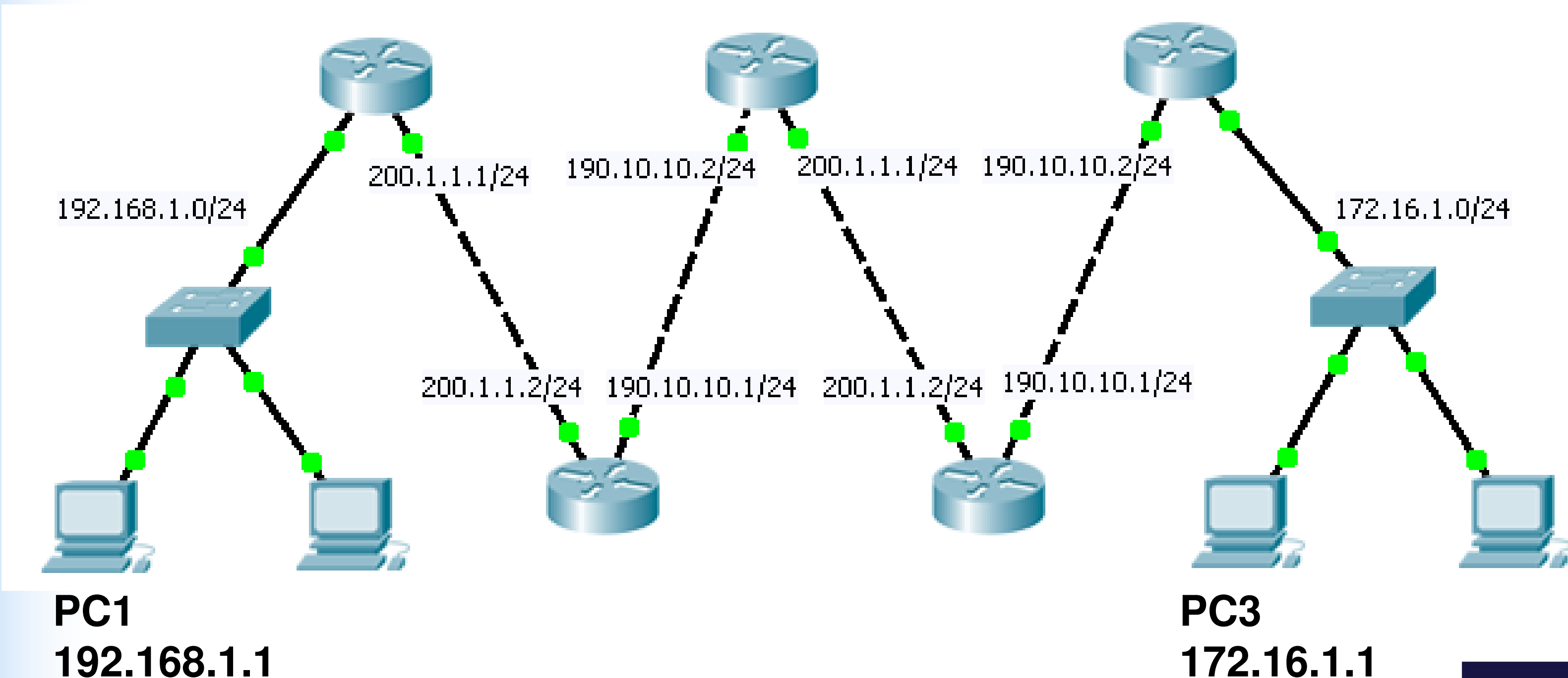
- Every IP address within the "closed" network has to be unique.
  - There CAN'T be two same IP addresses in the subnet.
  - Same applies to the Public IP addresses in the Internet.

- Upgrade concept: IP addresses within the "closed" network has to be unique, especially stub network ("end" networks).
  - However, IP address for intermediate networks can be "re-used".
  - Which means IP address in the "closed" network ARE NOT necessarily unique for some design

- But why do you need to have two same IP addresses within a closed network?
  - To save the number of IP addresses used, especially in a large network.
  - Normally, it is not encouraged.

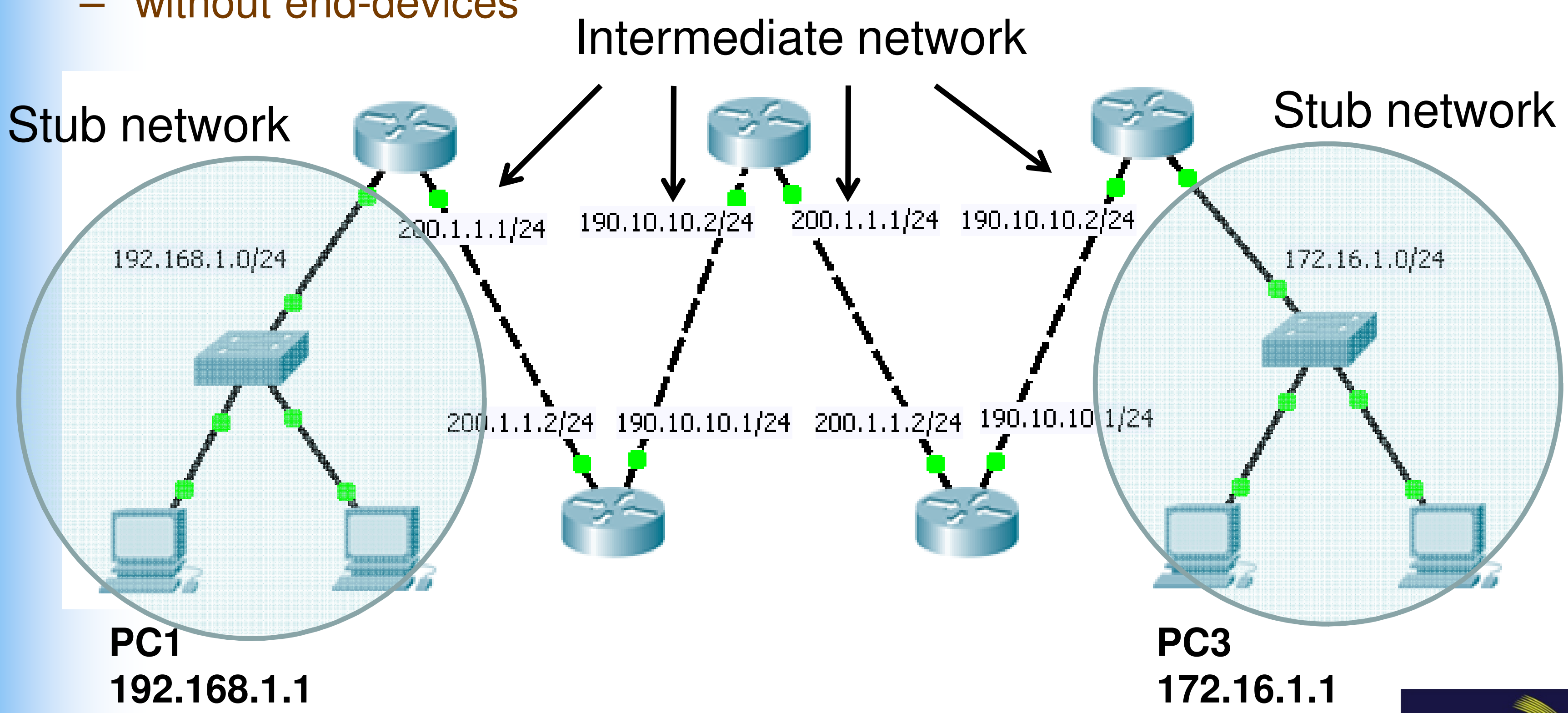
UCCN2243: IP Subnet Rule #1

- PC1 can ping PC3.
- Notice that there are “duplicated” IP addresses in the closed network. Where are the duplicated IP addresses?
- Quick quiz: which are the “stub networks” & which are the “intermediate network”?



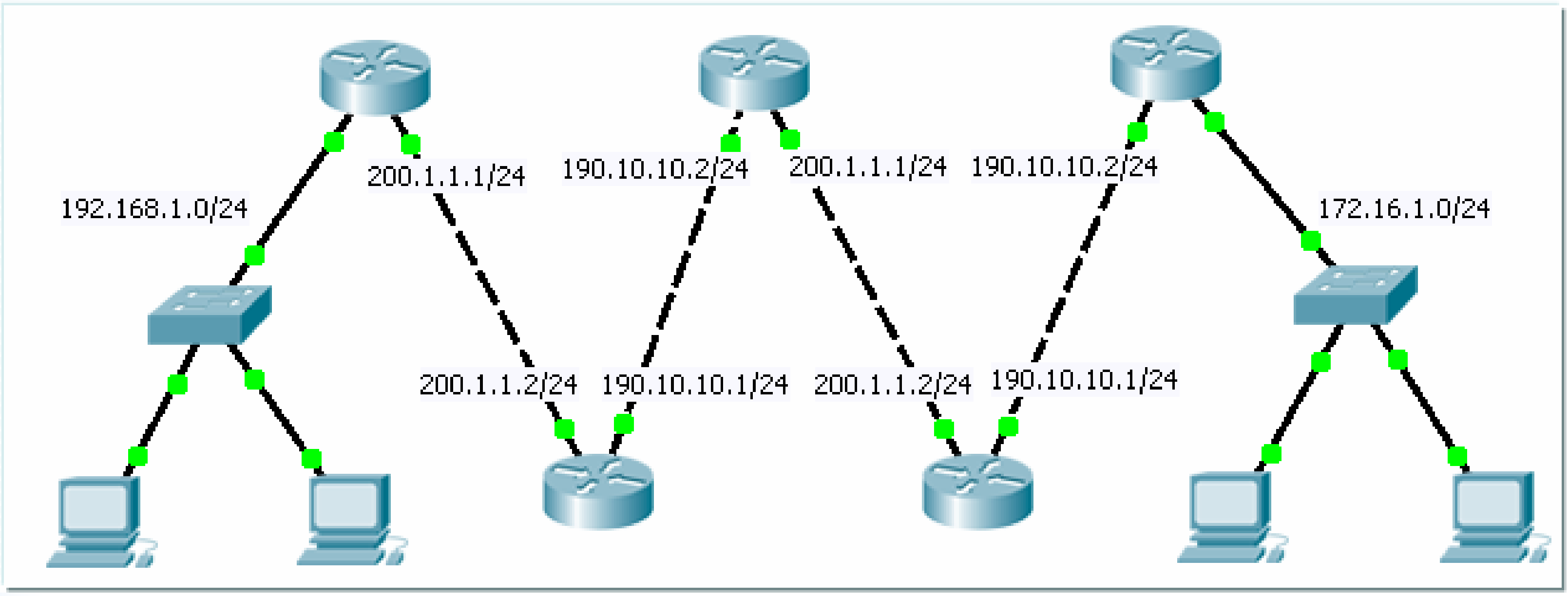
Answer

- Duplicate IP addresses
  - 200.1.1.1;   200.1.1.2; 190.10.10.1; 190.10.10.2;
- Subnets with end-devices are “stub network”
- Subnets between routers are “intermediate network”.
  - without end-devices



Quick Quiz: Non-unique IP schemes

- What is the pro and cons of the following network?



Answer

- Pros: IP address can be re-used in intermediate network as to “save” IP addresses (since IPv4 address is running out in the global Internet).

- Cons: You can’t access all of these non-unique IP address from a source.

IP Subnet Rule #2

1 IP subnet share 1 network ID

UCCN1004: IP Subnet Rule #2

- The function of a subnet mask is to divide the IP address into two parts: The Network ID and the Host ID.
  - The more important part is to produce the Network ID
- Subnet mask by itself is meaningless.
  - It has to "work" with an IP address.
- The process of getting the networking ID is to perform a bitwise AND operation between the IP and subnet mask.

Logical Bitwise Operations

bit 1	bit 2	OR (|)	AND (&)	XOR (^)
0	0	0	0	0
1	0	1	0	1
0	1	1	0	1
1	1	1	1	0

UCCN1004: IP Subnet Rule #2

- An example using an IP address of 156.154.81.56 used with a network mask of 255.255.255.240 follows:

```
IP Address:    10011100.10011010.01010001.00111000
Subnet mask:   11111111.11111111.11111111.11110000
Bitwise AND    -----------------------------------
Result:        10011100.10011010.01010001.00110000
```

- This translate to a network ID of 156.54.81.48
- Sometimes, <u>network ID</u> is also called <u>network address</u> or <u>subnet address</u>

UCCN2243: IP Subnet Rule #2

- Network ID is a very important concept.
  - Network ID forms the major component in the routing table of a router
- Host ID is less important.
  - However, the number of bits of the host ID can tell the network designer on how many hosts that he/she can put into that particular subnet.
- Every end device within the same subnet should have the same subnet mask as a good design practice.

UCCN2243: IP Subnet Rule #2

- 1 IP subnet must share 1 same network ID.

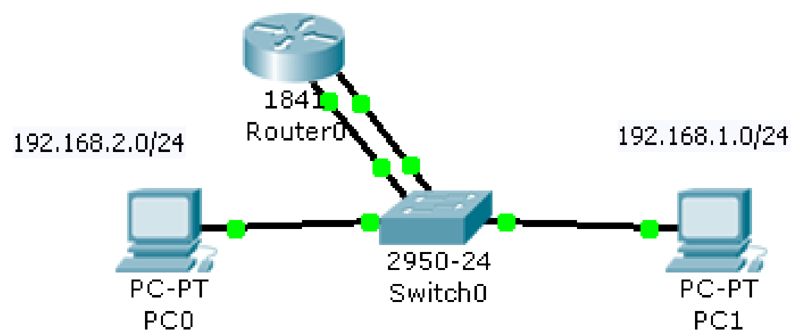
- Good design practice (important !!):
  - 1 broadcast domain = 1 IP subnet

UCCN2243: IP Subnet Rule #2

- UCCN1004 definition of LAN
  - End devices connected to a switch/hub/access point, bound by routers
- Upgrade: This is a "topological LAN"
- VLAN is a "logical LAN" which form a broadcast domain within a "topological LAN".
  - Condition: topological LAN must have a managed switch
- Broadcast domain is a logical region in a topological LAN, in which all IP interfaces that connect to this "region" receive the same broadcast frame.

UCCN2243: Quiz on Broadcast Domain

- How many IP subnets and broadcast domain are there in this network, if
  - A: PC0 and PC1 belong to the same VLAN, and there is only 1 VLAN in Switch0
  - B: PC0 and PC1 belong to different VLAN, and there are 2 VLAN in Switch0

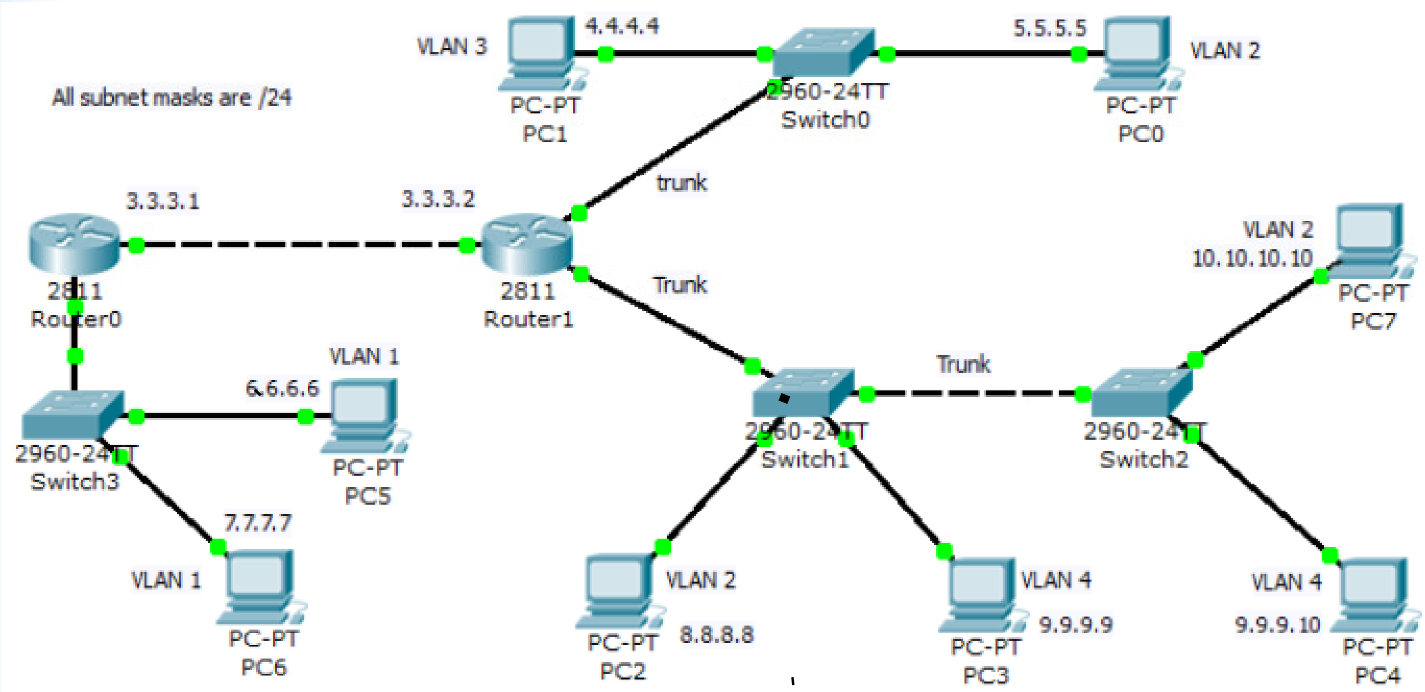


Answer

- A: 2 subnets, 1 broadcast domain
- B: 2 subnets, 2 broadcast domain
- Upgrade your “concept”:
  - Under “well behave” circumstances, broadcast domain = IP subnets.
  - Broadcast domain is a “hardware” mechanism, done by the switch.

UCCN2243: IP Subnet Rule #2

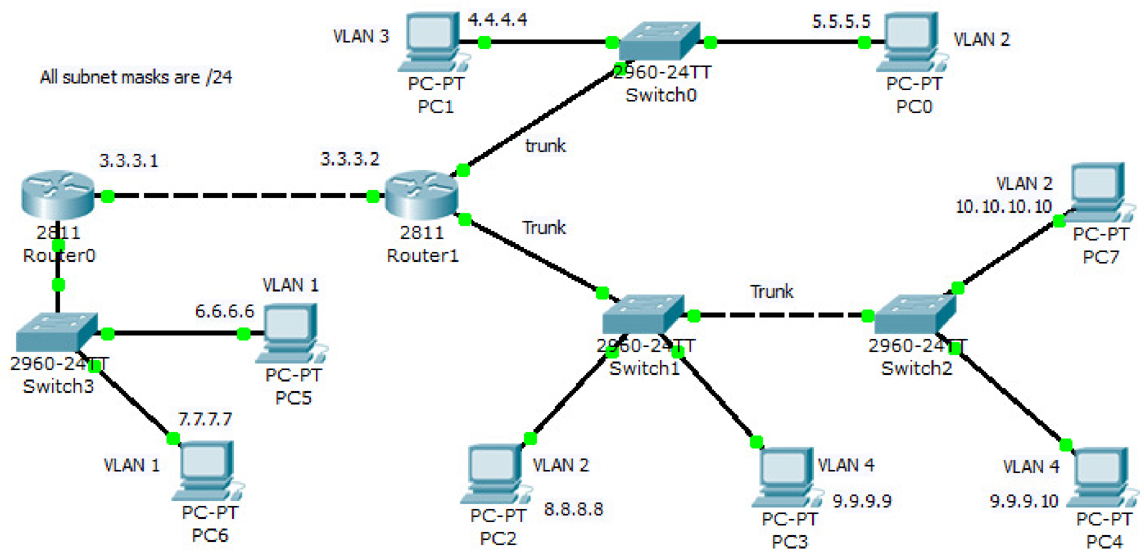
- Quiz: How many () in this network ?
  - Topological LAN
  - VLAN
  - Broadcast domain
  - IP subnet



Answer

- Quiz: How many () in this network ?
  - Topological LAN = 4
  - VLAN = 5
  - Broadcast domain = 6
  - IP subnet = 8

- Can you figure out the answer?



UCCN2243 Rule 2 Summary

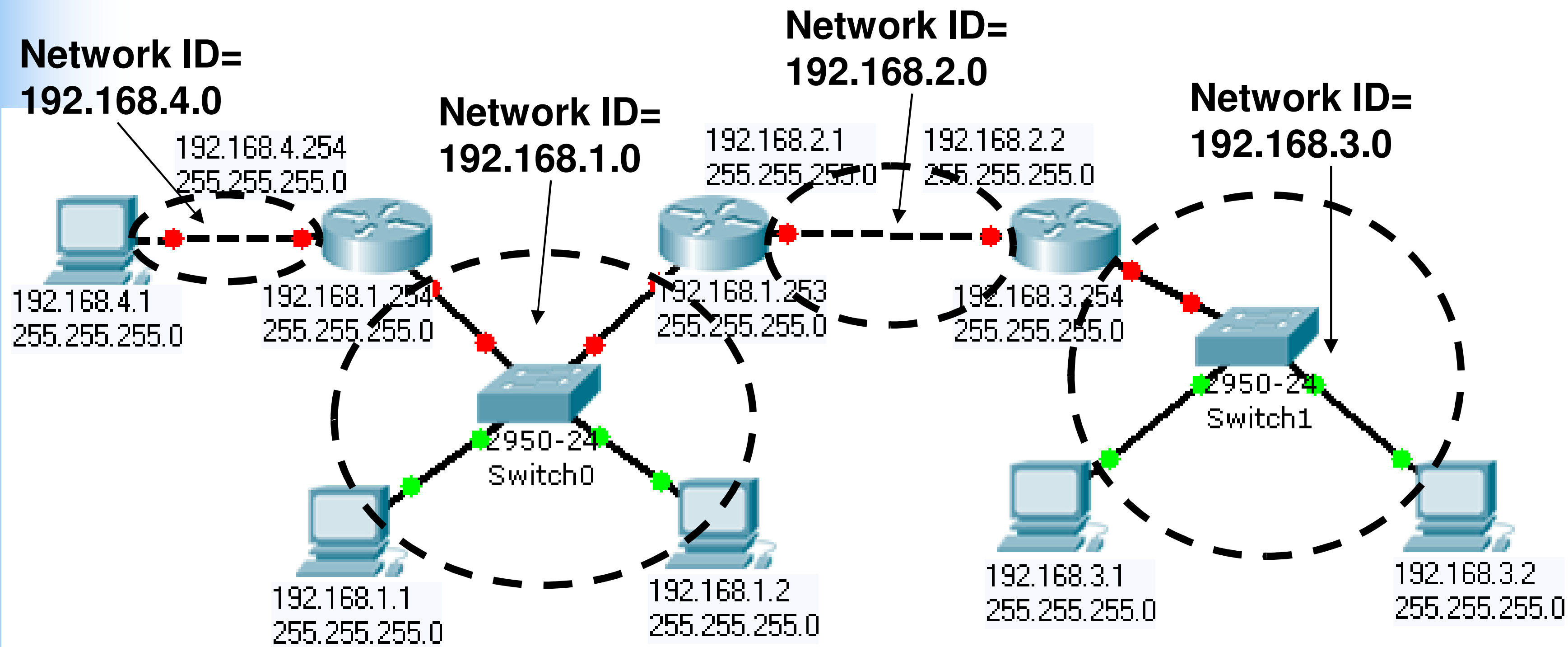
- IP subnet definition = IP interfaces that share the same network ID in a broadcast domain.
- 1 topological LAN = 1 broadcast domain (if there is only 1 VLAN in the topological LAN)
- 1 VLAN = 1 broadcast domain.
- Important and good design practice:
  - 1 broadcast domain = 1 IP subnet
  - Every IP interface within the same broadcast domain should have the same subnet mask.
    - Because this will lead to the same network ID

IP Subnet Rule #3

Does the network ID of a closed network have to unique?

UCCN1004: IP Subnet Rule #3

- Network address of the subnet within a "closed" network is preferred to be unique.

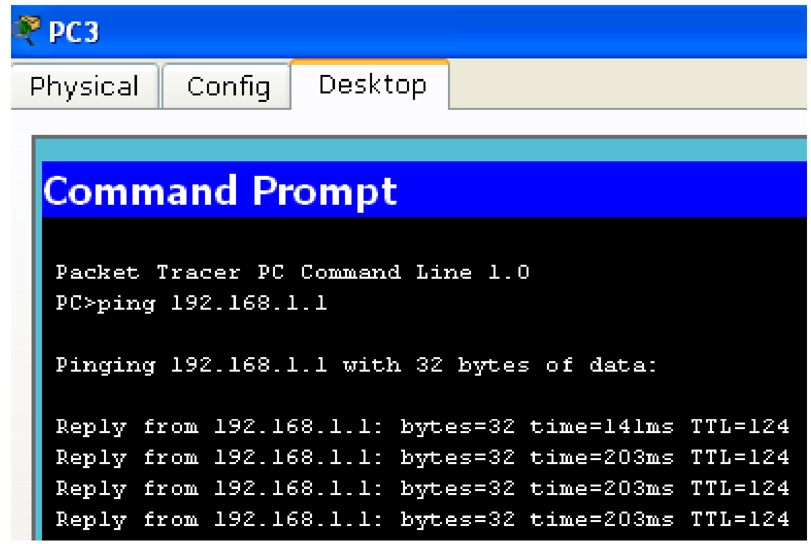


UCCN2243: IP Subnet Rule #3

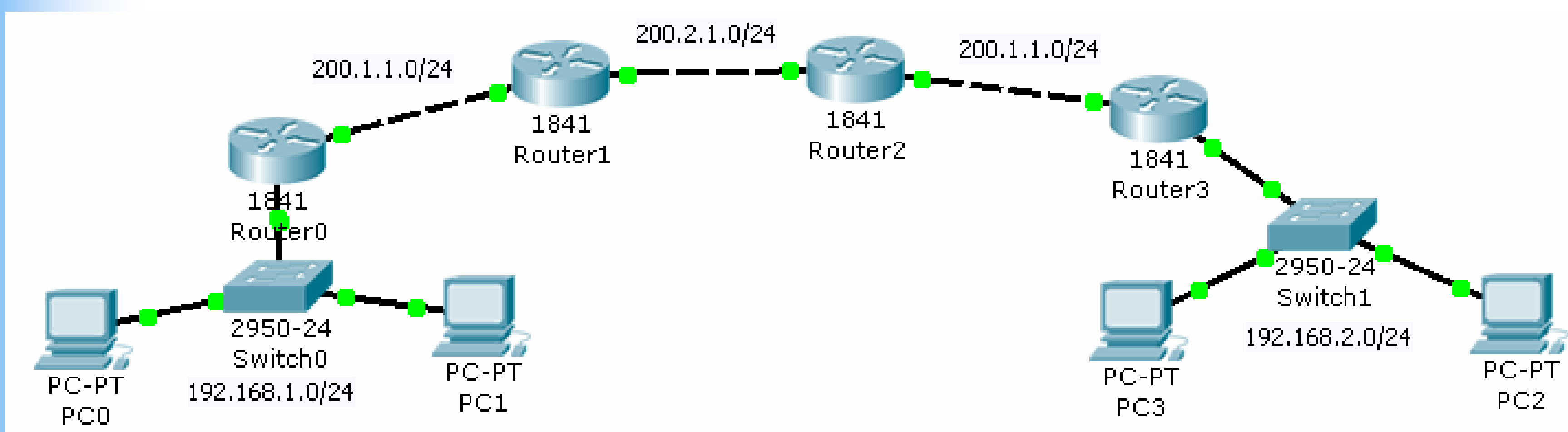
- After applying IP Subnet Rule #3, Network ID of a subnet within a “closed” network is preferred to be unique.
  - Preferred only, it does not say that it has to be necessary.
- However, in case where “IP addresses” are “not enough”, the network ID of the intermediate point-to-point network between routers can be “re-use”.
  - But use it sparingly, and carefully.
  - If you can avoid this, please avoid.
  - Demonstrated in “Subnet Rule #1”
- The network ID of the “stub subnets” with PC hosts is preferred to be unique by design.

UCCN2243: IP Subnet Rule #3

- The follow example shows that PC3 can ping PC0, though there are two intermediate network with the same network ID.

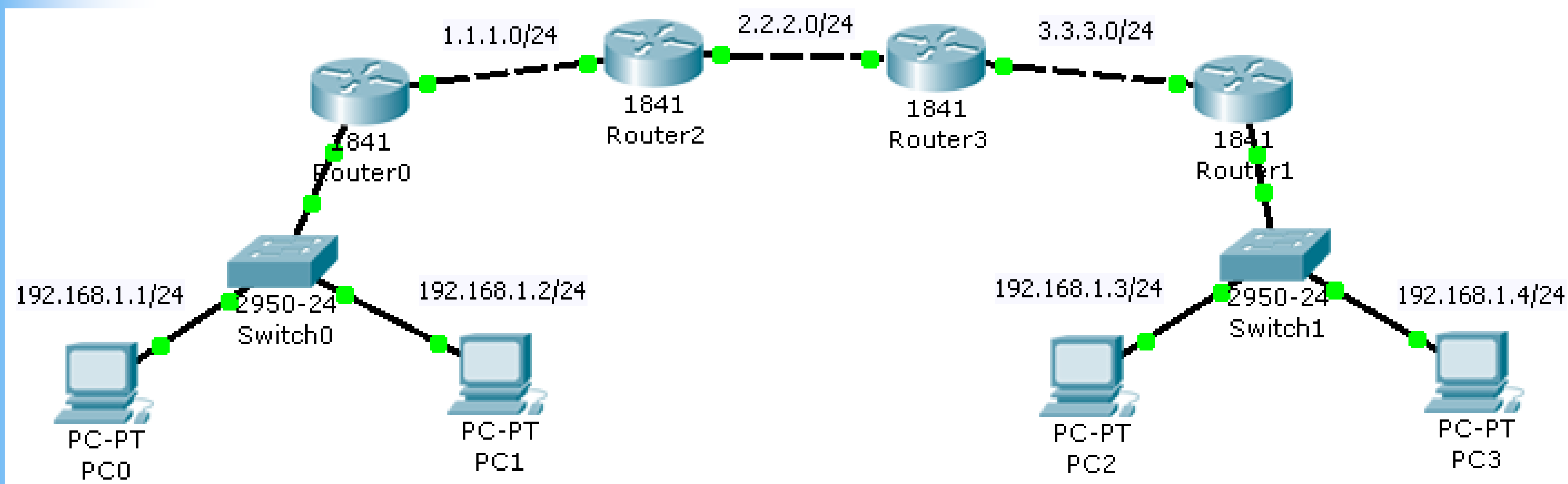


- However, PC3 can't ping all the IP addresses in this network.
  - Guess which IP addresses?



Quick Quiz : IP Subnet Rule #3

- What is the problem with the following network?



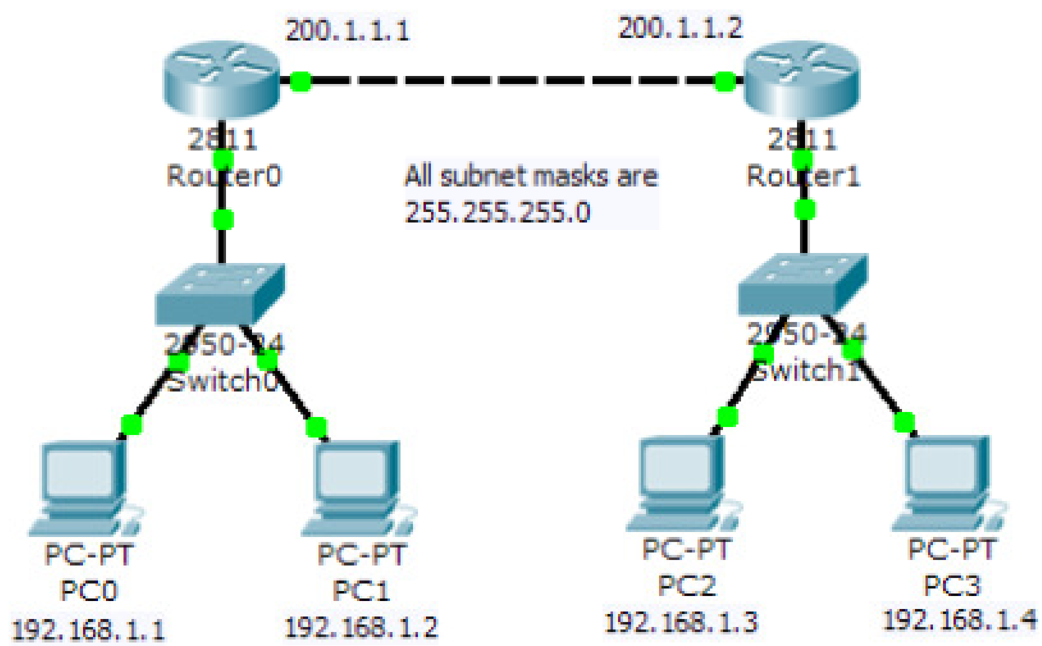
Answer

- This is a network with unique IP addresses but with two subnets with same network ID.
- PC0 and PC1 can never reach PC2 and PC3.

- Note: Routing tables will only register 1 unique destination network.
  – Other duplicate destination network will not be in the routing table.

UCCN1004: Packet Tracer Example

- In the following network, though every IP address is unique, but the network ID is not unique, hence the following network won't work.
  - You may try to build this network in Packet Tracer.



IP Subnet Rule #4

Slash notation of subnet mask as a better design parameters compared to dot notation?

UCCN1004: IP Subnet Rule #4

- For a correct and valid 32 bits subnet mask: Left all ‘1’ and right all ‘0’
  - 1111111.11110000.00000000.00000000 (valid)
  - 11111111.11101101.00000000.00000000 (not valid)
    - Subnet mask can’t have a ‘0’ between two ‘1’s or a ‘1’ between two ‘0’
- There are only 32 valid subnet masks (theoretical maximum).
  - 255.0.0.0, 255.128.0.0, 255.192.0.0, ……. 255.255.255.255
- Can be represented by ‘/’ notation
  - e.g. /9, /24, etc
  - /10 means ten ‘1’ from the left, and the remaining 22 bits are ‘0’
- 192.168.1.15/24 =>
  - This interface has an IP = 192.168.1.15
  - The subnet mask = 255.255.255.0
  - Belongs to network 192.168.1.0

UCCN2243: IP Subnet Rule #4

- Similar to Rule #2, slash notation can be a better metric for the subnet design in terms of maximum allowable number of hosts in the subnet.
- Quick Quiz: What is the maximum number of hosts available with a subnet mask of /27?

Answer

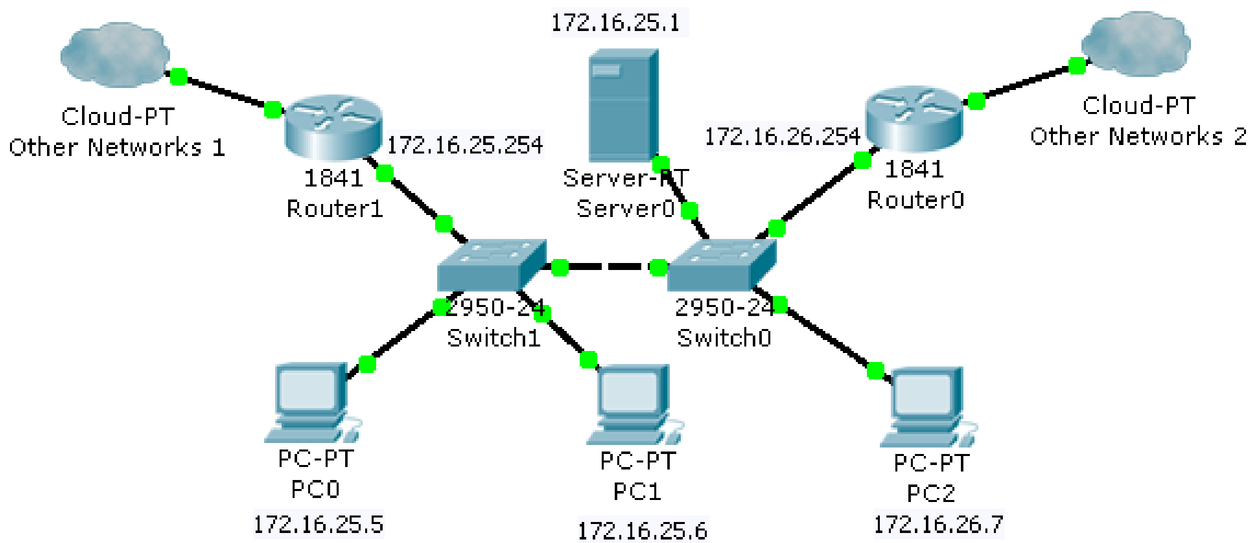
- /27 = 27 bits of network ID
- Host bits = 32 bits – 27 bits = 5 bits
- Maximum allowable # of hosts = $2^5 - 2 = 30$

IP Subnet Rule #5

Within a WAN subnet, does the IP address of all hosts and gateways have the same network ID?

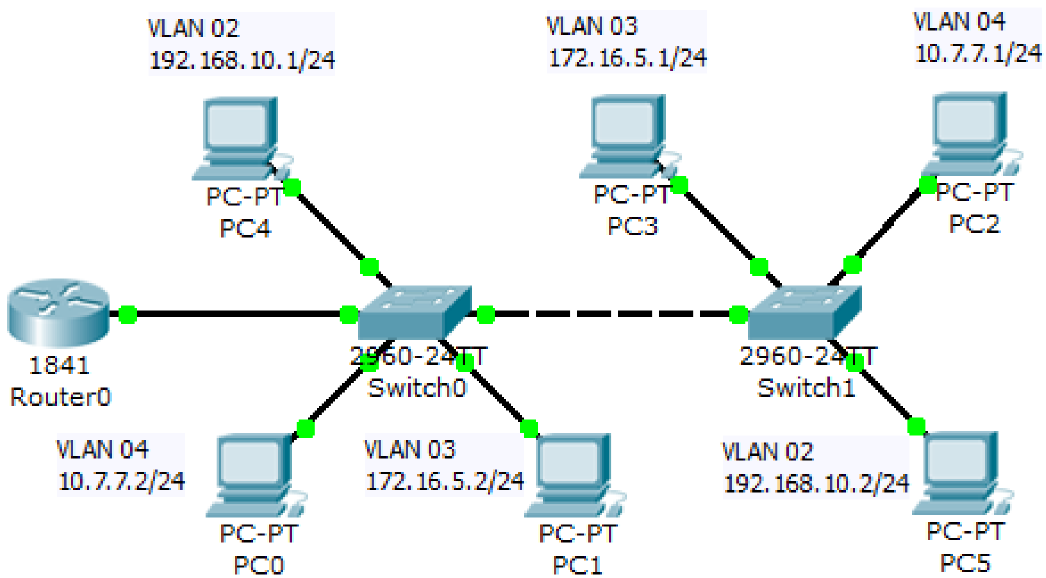
UCCN1004: IP Subnet Rule #5

- In order to communicate within the "LAN", the IP address of all hosts and gateways must have the same network ID.
  - All IP in the same subnet should be designed to have the same network ID.
    - That includes PC IP, server IP, and gateway IP.
- For the following example:
  - If subnet mask = /16, all PC IP, server0 IP, and gateway IP will be able to communicate with each other
  - If subnet mask = /24, gateway IP of Router0, and PC2 WON'T be able to communicate with the rest.



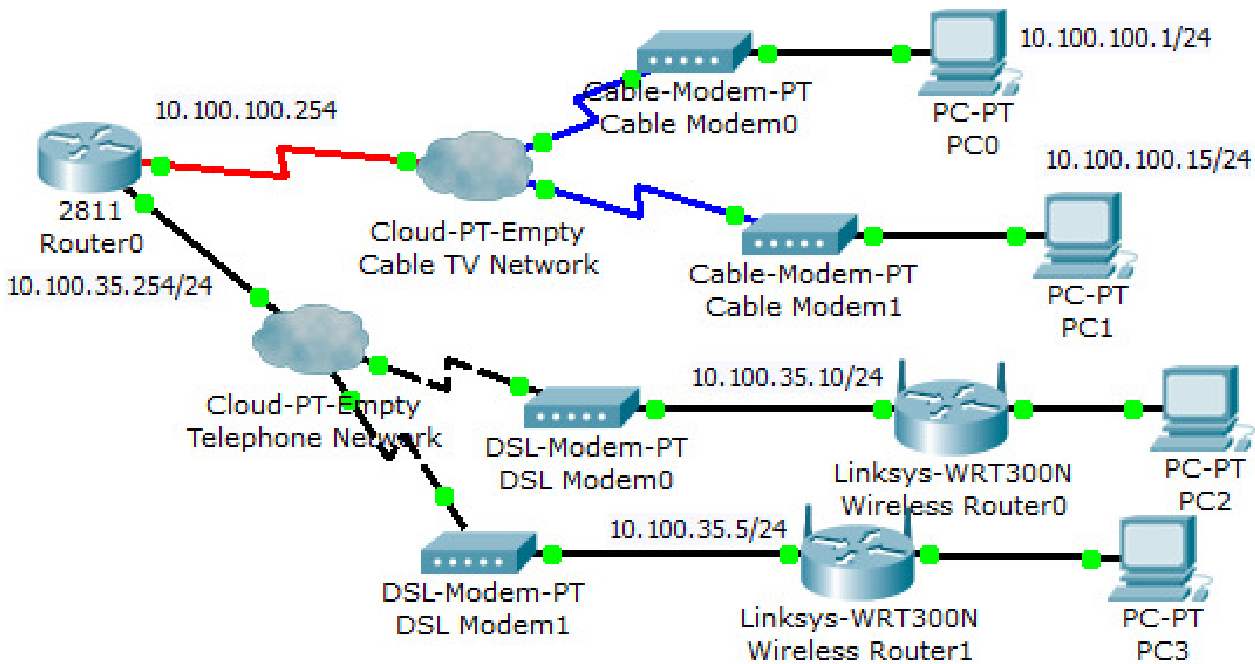
UCCN2243: IP Subnet Rule #5

- In order to communicate within the "virtual LAN", the IP address of all hosts and gateways must have the same network ID.
  - All IP in the same subnet of a virtual LAN should be designed to have the same network ID.
    - That includes PC IP, server IP, and gateway IP.
- For the following example:
  - VLAN 02 has the network ID of 192.168.10.0
  - VLAN 03 has the network ID of 172.16.5.0
  - VLAN 04 has the network ID of 10.7.7.0



UCCN2243: IP Subnet rule #5

- In order to communicate within the "WAN", the IP address of all hosts and gateways must have the same network ID.
  - Rule #5 applies in all LAN, VLAN and WAN (wide area network)
- For the following example:
  - Cable TV network has a network ID of 10.100.100.0
  - Telephone network has a network ID of 10.100.35.0

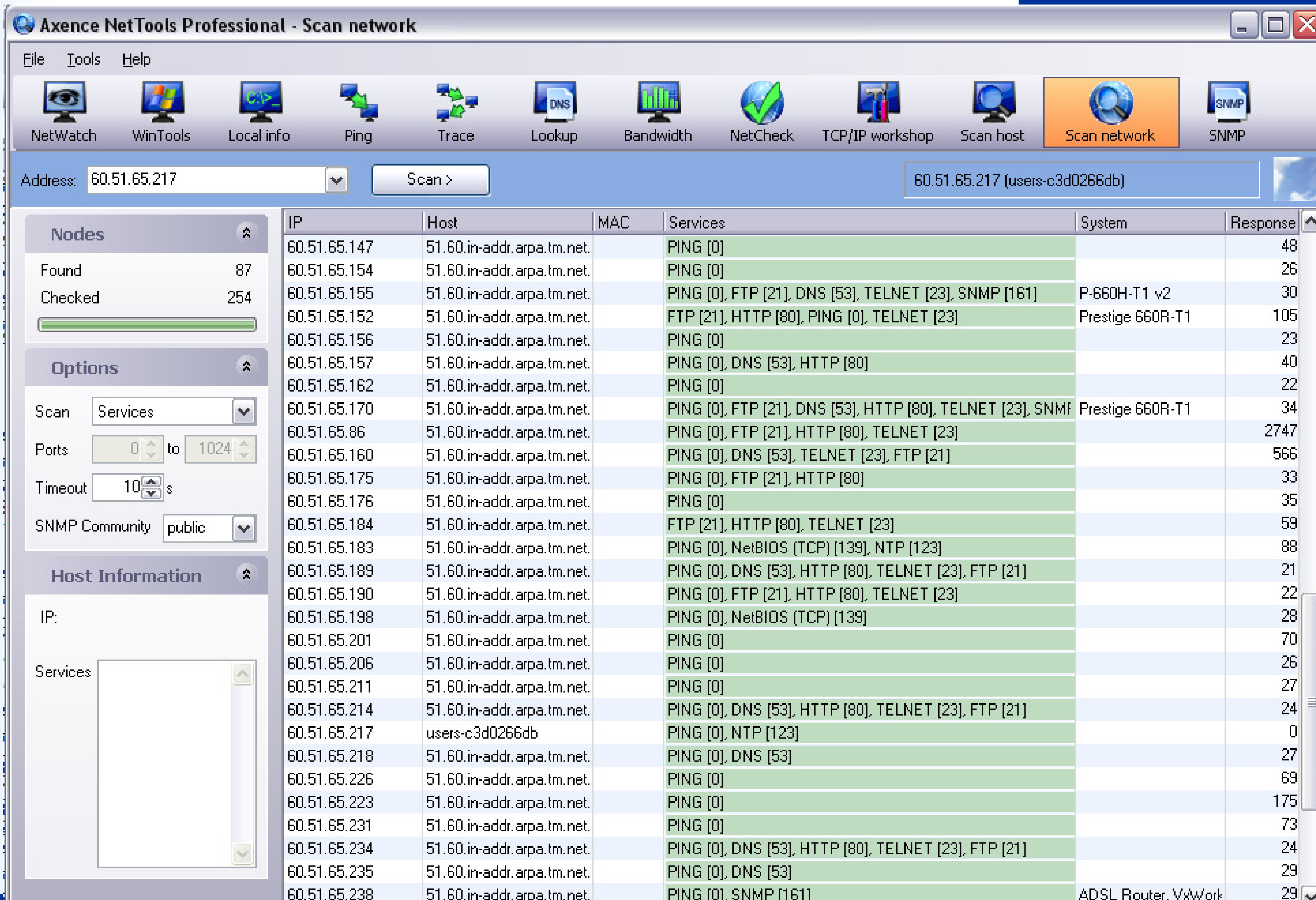


UCCN2243: Scanning Streamyx Subnet

- Your Streamyx line connects to a WAN.
- You can verify that Streamyx IP is part of a bigger subnet.
- First, you check your Streamyx public using “ipconfig”
- Next, use a Net Tool to scan the subnet to check for other “members” of the same subnet.

```
C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\User>ipconfig

Windows IP Configuration


Ethernet adapter Local Area Connection 3:

        Connection-specific DNS Suffix  . :
        IP Address. . . . . . . . . . . . : 192.168.1.3
        Subnet Mask . . . . . . . . . . . : 255.255.255.0
        Default Gateway . . . . . . . . . : 192.168.1.1

PPP adapter Streamyx:

        Connection-specific DNS Suffix  . :
        IP Address. . . . . . . . . . . . : 60.51.65.217
        Subnet Mask . . . . . . . . . . . : 255.255.255.255
        Default Gateway . . . . . . . . . : 60.51.65.217
```

UCCN2243: Scanning Streamyx Subnet



IP	Host	MAC	Services	System	Response
60.51.65.147	51.60.in-addr.arpa.tm.net.		PING [0]		48
60.51.65.154	51.60.in-addr.arpa.tm.net.		PING [0]		26
60.51.65.155	51.60.in-addr.arpa.tm.net.		PING [0], FTP [21], DNS [53], TELNET [23], SNMP [161]	P-660H-T1 v2	30
60.51.65.152	51.60.in-addr.arpa.tm.net.		FTP [21], HTTP [80], PING [0], TELNET [23]	Prestige 660R-T1	105
60.51.65.156	51.60.in-addr.arpa.tm.net.		PING [0]		23
60.51.65.157	51.60.in-addr.arpa.tm.net.		PING [0], DNS [53], HTTP [80]		40
60.51.65.162	51.60.in-addr.arpa.tm.net.		PING [0]		22
60.51.65.170	51.60.in-addr.arpa.tm.net.		PING [0], FTP [21], DNS [53], HTTP [80], TELNET [23], SNMI	Prestige 660R-T1	34
60.51.65.86	51.60.in-addr.arpa.tm.net.		PING [0], FTP [21], HTTP [80], TELNET [23]		2747
60.51.65.160	51.60.in-addr.arpa.tm.net.		PING [0], DNS [53], TELNET [23], FTP [21]		566
60.51.65.175	51.60.in-addr.arpa.tm.net.		PING [0], FTP [21], HTTP [80]		33
60.51.65.176	51.60.in-addr.arpa.tm.net.		PING [0]		35
60.51.65.184	51.60.in-addr.arpa.tm.net.		FTP [21], HTTP [80], TELNET [23]		59
60.51.65.183	51.60.in-addr.arpa.tm.net.		PING [0], NetBIOS (TCP) [139], NTP [123]		88
60.51.65.189	51.60.in-addr.arpa.tm.net.		PING [0], DNS [53], HTTP [80], TELNET [23], FTP [21]		21
60.51.65.190	51.60.in-addr.arpa.tm.net.		PING [0], FTP [21], HTTP [80], TELNET [23]		22
60.51.65.198	51.60.in-addr.arpa.tm.net.		PING [0], NetBIOS (TCP) [139]		28
60.51.65.201	51.60.in-addr.arpa.tm.net.		PING [0]		70
60.51.65.206	51.60.in-addr.arpa.tm.net.		PING [0]		26
60.51.65.211	51.60.in-addr.arpa.tm.net.		PING [0]		27
60.51.65.214	51.60.in-addr.arpa.tm.net.		PING [0], DNS [53], HTTP [80], TELNET [23], FTP [21]		24
60.51.65.217	users-c3d0266db		PING [0], NTP [123]		0
60.51.65.218	51.60.in-addr.arpa.tm.net.		PING [0], DNS [53]		27
60.51.65.226	51.60.in-addr.arpa.tm.net.		PING [0]		69
60.51.65.223	51.60.in-addr.arpa.tm.net.		PING [0]		175
60.51.65.231	51.60.in-addr.arpa.tm.net.		PING [0]		73
60.51.65.234	51.60.in-addr.arpa.tm.net.		PING [0], DNS [53], HTTP [80], TELNET [23], FTP [21]		24
60.51.65.235	51.60.in-addr.arpa.tm.net.		PING [0], DNS [53]		29
60.51.65.238	51.60.in-addr.arpa.tm.net.		PING [0], SNMP [161]	ADSL Router, VxWork	29

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Study Internet as "IP subnets"

Visual Trace Route Tool

approximate geophysical trace

trace the path to a network

Remote Address india.gov.in Host Trace Proxy Trace

Use Current IP

Host Trace

yougetsignal.com → Remote Address

trace information

Host trace to india.gov.in
21 hops / 16.2 seconds

1. dreamhost.com
2. dreamhost.com
3. pnap.net
4. pnap.net
5. ALTER.NET
6. ALTER.NET
7. ALTER.NET
8. ALTER.NET
9. 204.255.169.90
10. as6453.net
11. as6453.net
12. as6453.net
13. as6453.net
14. as6453.net
15. vsnl.net.in
16. vsnl.net.in
17. 172.31.128.138
18. 172.31.128.138
19. vsnl.net.in
20. 164.100.193.5
21. Unknown

~12,391 miles traveled

Redraw Trace

Data travels
subnet by
subnet
via the Internet

IP Subnet Rule #6

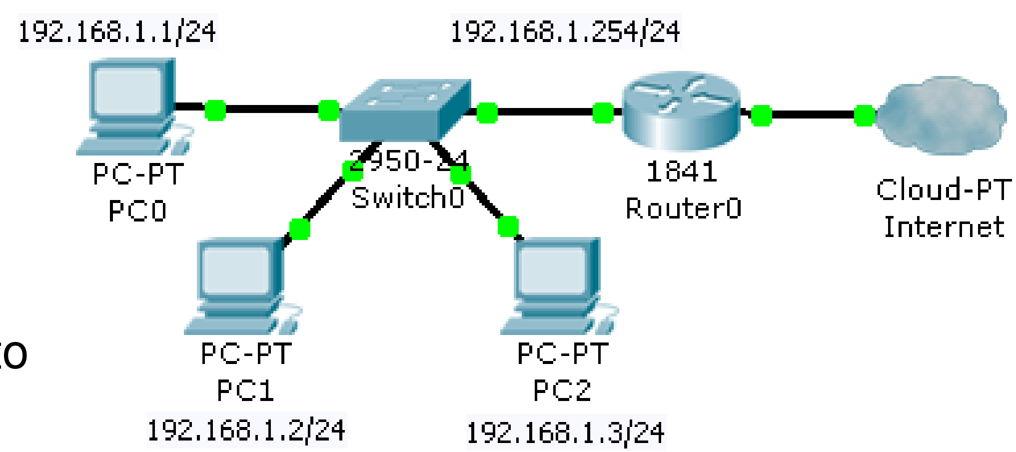
The issues of gateway in an IP subnet.

UCCN1004: IP Subnet Rule #6

- If the data's destination IP does not have the same network ID as the source IP, the data have to be sent to the gateway (router).
- In most cases under Windows, the data with destination IP that does not have the same network ID will be sent to default gateway.
  - The default gateway IP will be utilized in ARP.
  - Destination MAC address of default gateway will be used.

If PC0 wants to:
ping 192.168.2.1
ping 180.7.4.3
ping 10.0.1.1

The data has to be sent to the gateway

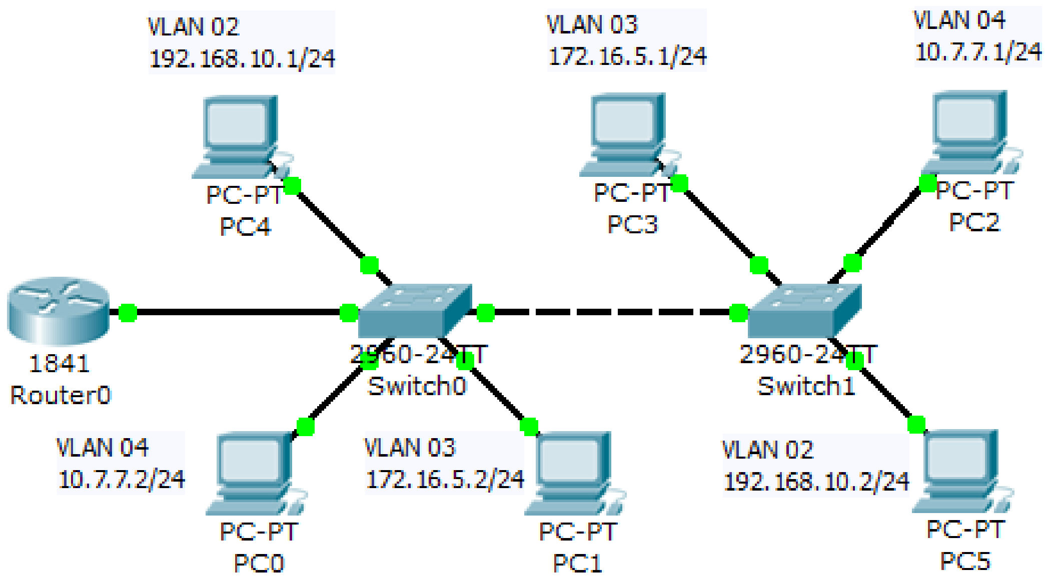


UCCN2243: IP Subnet Rule #6

- If the data's destination IP does not have the same network ID as the source IP, the data have to be sent to the gateway (router).

- Same rule apples to VLAN.

- Quick Quiz: For the following network:
  - How many gateway IP are required?
  - What is/are the gateway IP(s) if the gateway IP is set as the last usable IP?



Answer

- 3 VLANs => 3 gateway IP are required
- Gateway IP
  - VLAN 02: 192.168.10.254
  - VLAN 03: 172.16.5.254
  - VLAN 04: 10.7.7.254

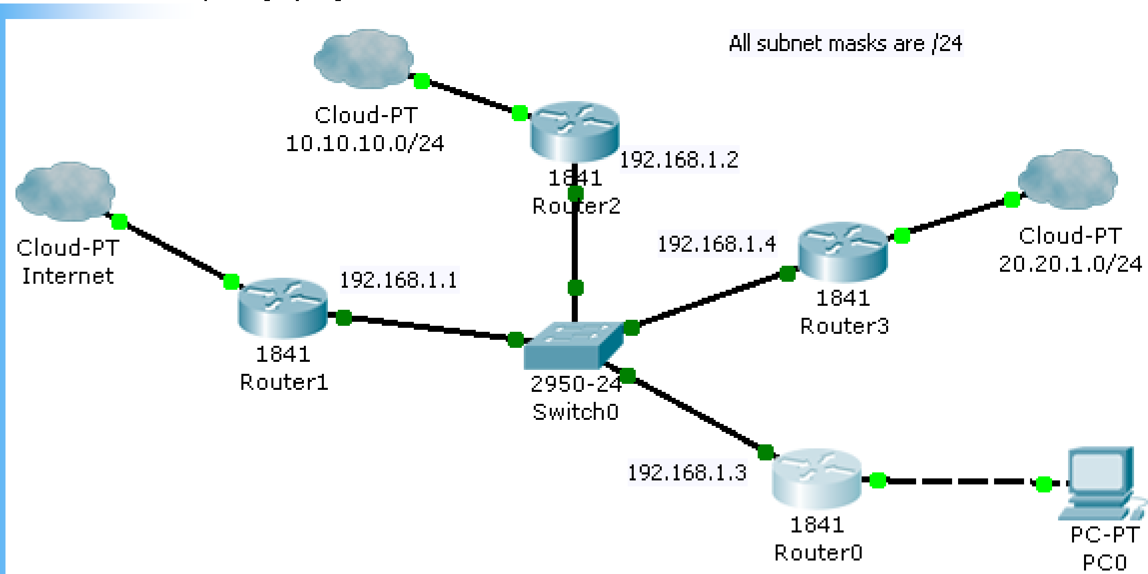
UCCN2243: IP Subnet Rule #6

- Upgrade of this rule: Before the packet is sent to a gateway, it will first check the routing table, whether it is in a router or in a PC.
- Even your PC has a routing table.

```
    > route PRINT

C:\Documents and Settings\User>route PRINT
===========================================================================
Interface List
0x1 ........................... MS TCP Loopback interface
0x2 ...00 1f d0 d1 36 40 ...... Realtek RTL8168C(P)/8111C(P) PCI-E Gigabit Ether
net NIC - Packet Scheduler Miniport
===========================================================================
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      192.168.1.1     192.168.1.3     20
        127.0.0.0        255.0.0.0        127.0.0.1       127.0.0.1      1
      192.168.1.0    255.255.255.0      192.168.1.3     192.168.1.3     20
      192.168.1.3  255.255.255.255        127.0.0.1       127.0.0.1     20
    192.168.1.255  255.255.255.255      192.168.1.3     192.168.1.3     20
        224.0.0.0        240.0.0.0      192.168.1.3     192.168.1.3     20
  255.255.255.255  255.255.255.255      192.168.1.3     192.168.1.3      1
Default Gateway:       192.168.1.1
===========================================================================
Persistent Routes:
  None

C:\Documents and Settings\User>
```

Quick Quiz: IP Subnet Rule #6

```
Gateway of last resort is 192.168.1.2 to network 0.0.0.0

     9.0.0.0/24 is subnetted, 1 subnets
S       9.9.9.0 [1/0] via 192.168.1.2
     10.0.0.0/24 is subnetted, 1 subnets
S       10.10.10.0 [1/0] via 192.168.1.4
     12.0.0.0/24 is subnetted, 1 subnets
S       12.12.12.0 [1/0] via 192.168.1.4
     30.0.0.0/24 is subnetted, 1 subnets
S       30.30.30.0 [1/0] via 192.168.1.1
C    192.168.1.0/24 is directly connected, FastEthernet0/0
S*   0.0.0.0/0 [1/0] via 192.168.1.2
```



- Which gateway will it go to, if Router0 tries to ping the following IP addresses
  - 30.30.30.2
  - 10.10.10.1
  - 27.27.27.3

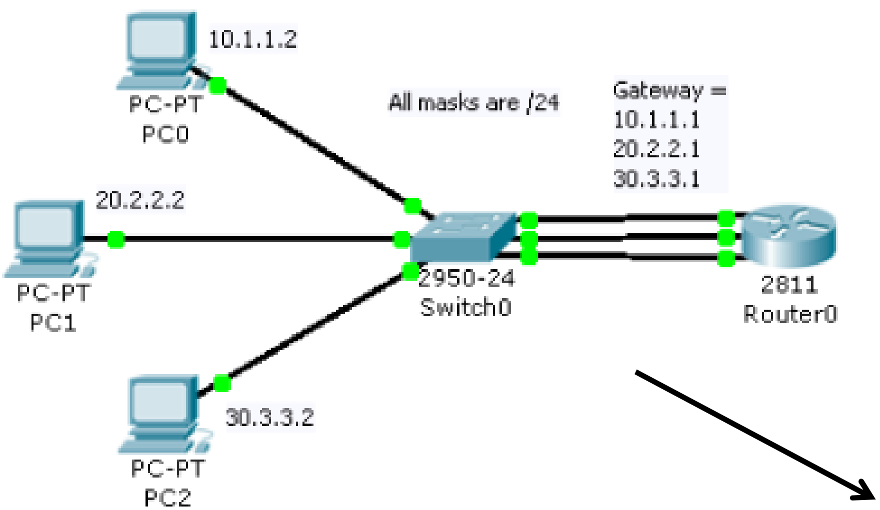
Answer

```
Gateway of last resort is 192.168.1.2 to network 0.0.0.0

     9.0.0.0/24 is subnetted, 1 subnets
S       9.9.9.0 [1/0] via 192.168.1.2
     10.0.0.0/24 is subnetted, 1 subnets
S       10.10.10.0 [1/0] via 192.168.1.4
     12.0.0.0/24 is subnetted, 1 subnets
S       12.12.12.0 [1/0] via 192.168.1.4
     30.0.0.0/24 is subnetted, 1 subnets
S       30.30.30.0 [1/0] via 192.168.1.1
C    192.168.1.0/24 is directly connected, FastEthernet0/0
S*   0.0.0.0/0 [1/0] via 192.168.1.2
```

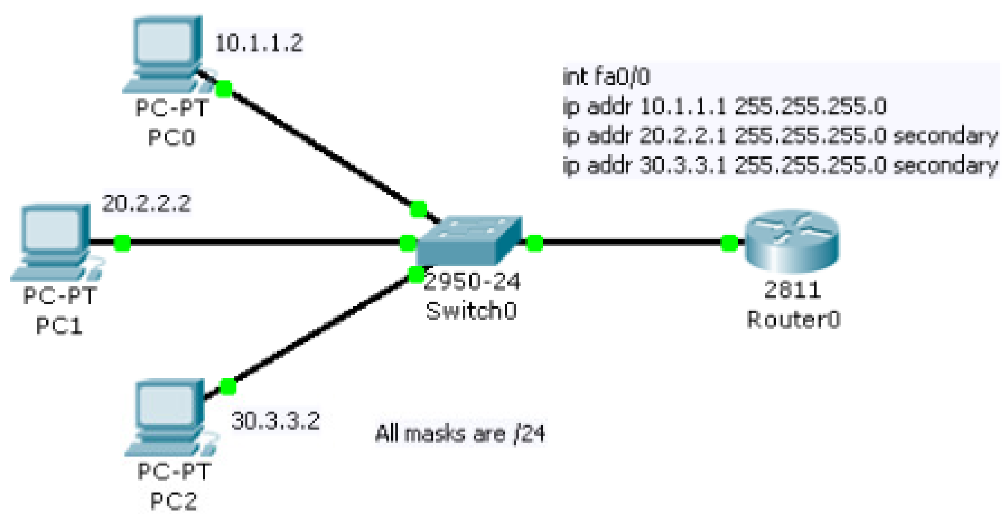
- 30.30.30.2
  - Gateway = (192.168.1.1)
- 10.10.10.1
  - Gateway = (192.168.1.4)
- 27.27.27.3
  - Gateway = (192.168.1.2)

IP address secondary

- We can put 3 IP address in the router interface with “ip address *IP mask* secondary”.



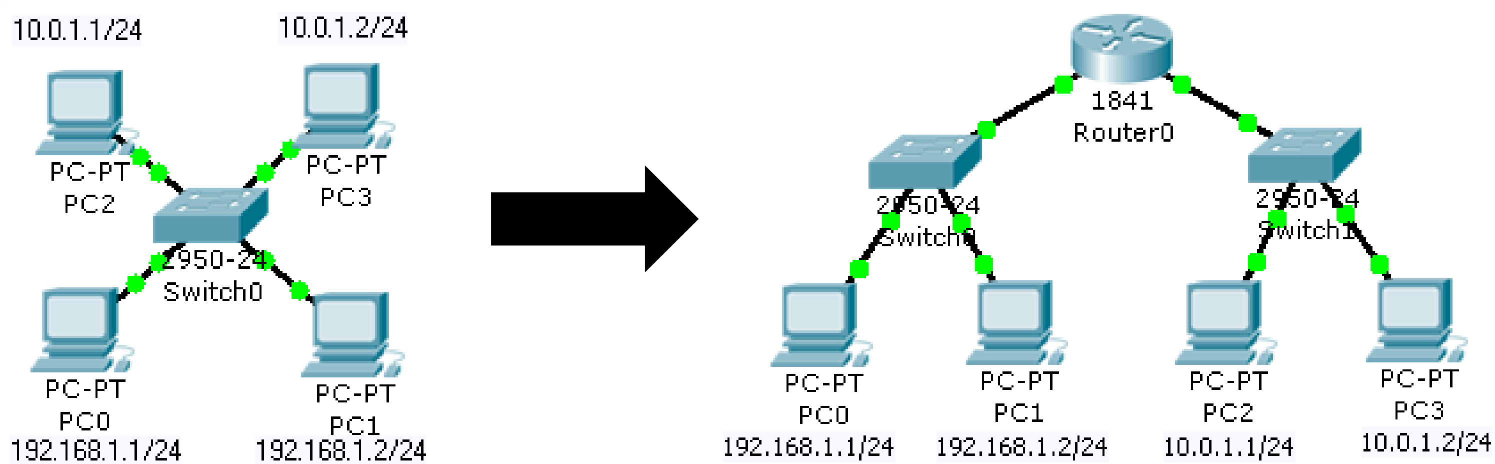
Note: supported by GNS3 and not packet tracer



IP Subnet Rule #7

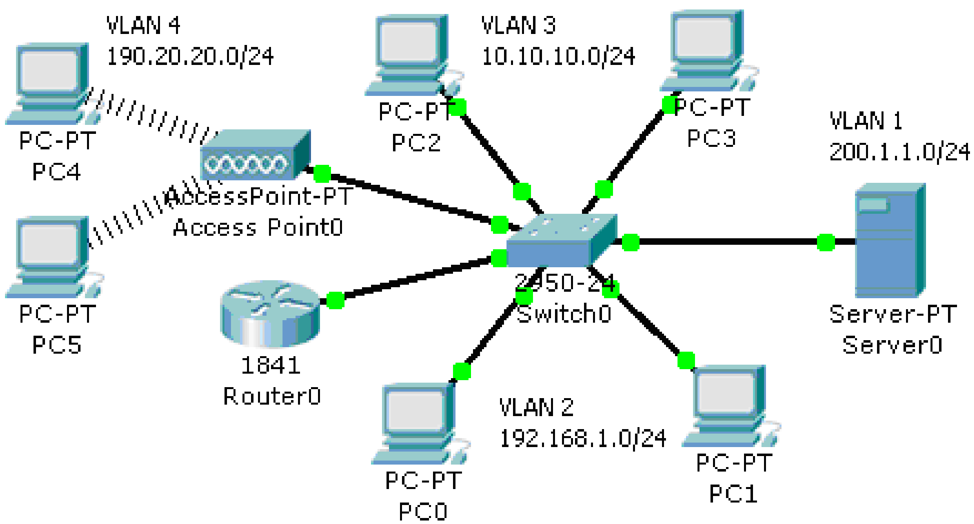
UCCN1004: IP Subnet Rule #7

- Router MUST be used in order for two hosts with different network address (or network ID) to communicate.
- Communication will not happen between hosts with different subnets address that are connected to a switch
- A switch only provides communication for the PCs with the same network ID
- Two different subnets has to be communicated via a router.



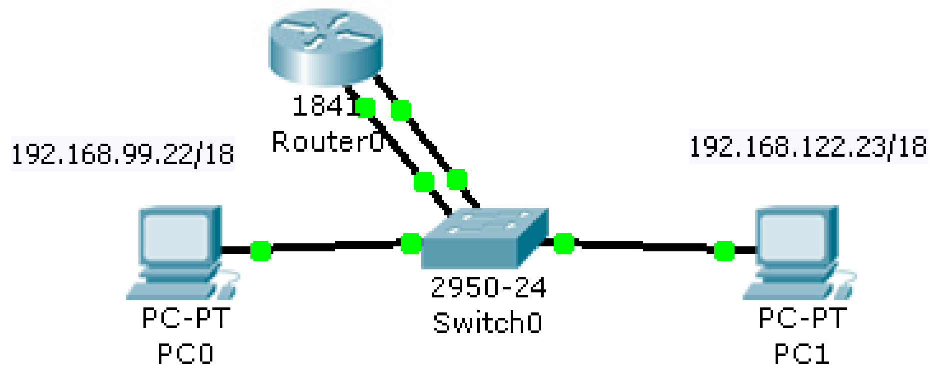
UCCN2243: IP Subnet rule #7

- Remember that VLAN does not violate this rule.
- Even though VLAN 3 and VLAN 4 are connected to the same switch but they can't communicate with each other.
- VLAN 3 and VLAN 4 require a router to form the communication between the these two VLAN
  - Remember sub-interfaces?



Quick Quiz: IP Subnet rule #7

- Do you need Router0 for the following network?



Answer:

- No, since both PCs have the same network ID.

IP Subnet Rule #8

UCCN1004: IP Subnet Rule #8

- Two special cases on host ID bits which are all ‘0’s and all ‘1’s
  - When Host ID bits are all ‘0’s, it is a network address.
  - When Host ID is all ‘1’s, it is a broadcast address, we don’t use it as a host address too.

- These two addresses represent the “head” and the “tail” of the given IP address block range.

- We CAN’T use both of these IP addresses as host IP and gateway IP.

- Example, IP address = 192.168.1.1, Subnet mask = 255.255.255.0
  - Network address of the subnet = 192.168.1.0
  - Broadcast address of the subnet = 192.168.1.255

IP Subnet Rule #9

UCCN1004: IP subnet rule #9

- The first usable IP and the last usable IP.
  - Usable IP addresses mean they can be used in hosts, PCs, printers, gateways, and servers.
- The first usable IP = network address + 1
  - More precisely, host ID = 1
  - If network adress = 192.168.3.0, first usable IP = 192.168.3.1
- The last usable IP = broadcast address – 1
  - More precisely, host ID = All ‘1’s – 1
  - If broadcast address = 192.168.3.255, last usable IP = 192.168.3.254
- Cisco guidelines (not rules):
  - Last usable IP is preferred to be
    - router IP address = gateway IP
  - First usable IP is preferred to be
    - server, printer
    - any host that requires static IP

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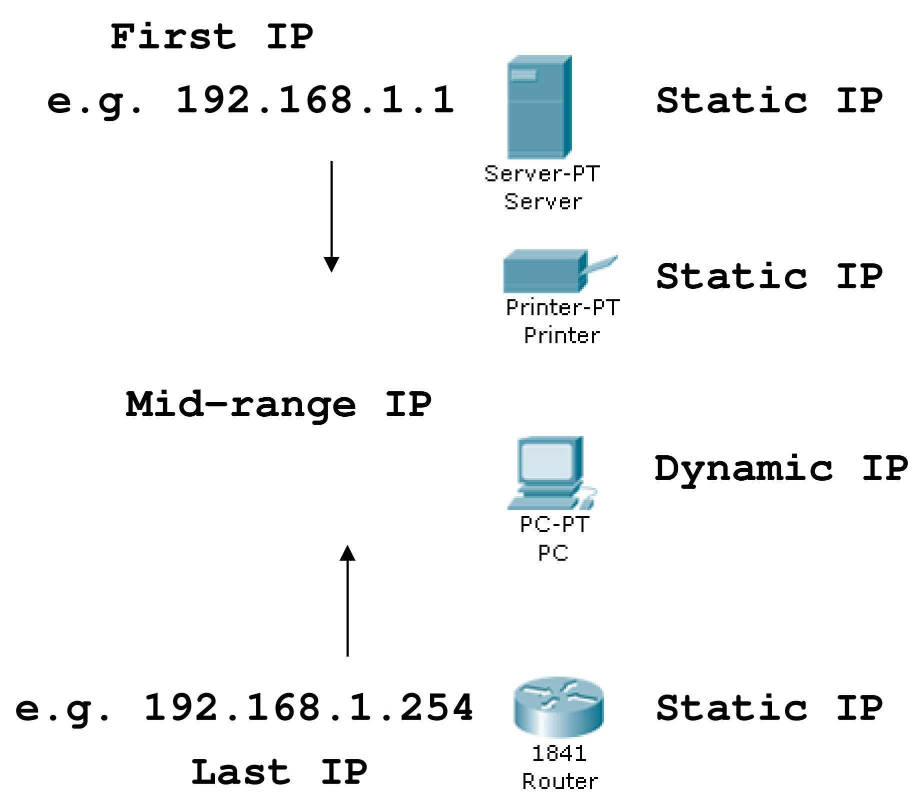
UCCN1004: LAN IP design guideline

- Given a range of IP address:
  - First IP addresses are preferred to be used for setting static IP for servers and printers
    - Starting from the first IP and counting down
    - e.g: 192.168.1.1 for DHCP server; 192.168.1.2 for printer
  - Last IP addresses are preferred to be used for setting router IP (gateway IP)
    - Starting from the last IP and counting up.
    - e.g. 192.168.1.254 for gateway 1, 192.168.1.253 for gateway 2.
  - Mid-range IP addresses are preferred to be set as the DHCP range for the PCs
    - e.g 192.168.1.10 to 192.168.1.250
    - In this range, we reserve 10 first IP for servers and printer and 5 last IPs for gateways



UCCN2243: Usable IP for Host (1)

- First and last usable IP show the usable range of the IP address.
- In a subnet, we want to have more IP address for hosts (PCs, workstations, servers, printers).
- Usable IP for router (gateway) can be considered as “wasted”.
  - Unless router is used for certain services such as DHCP.

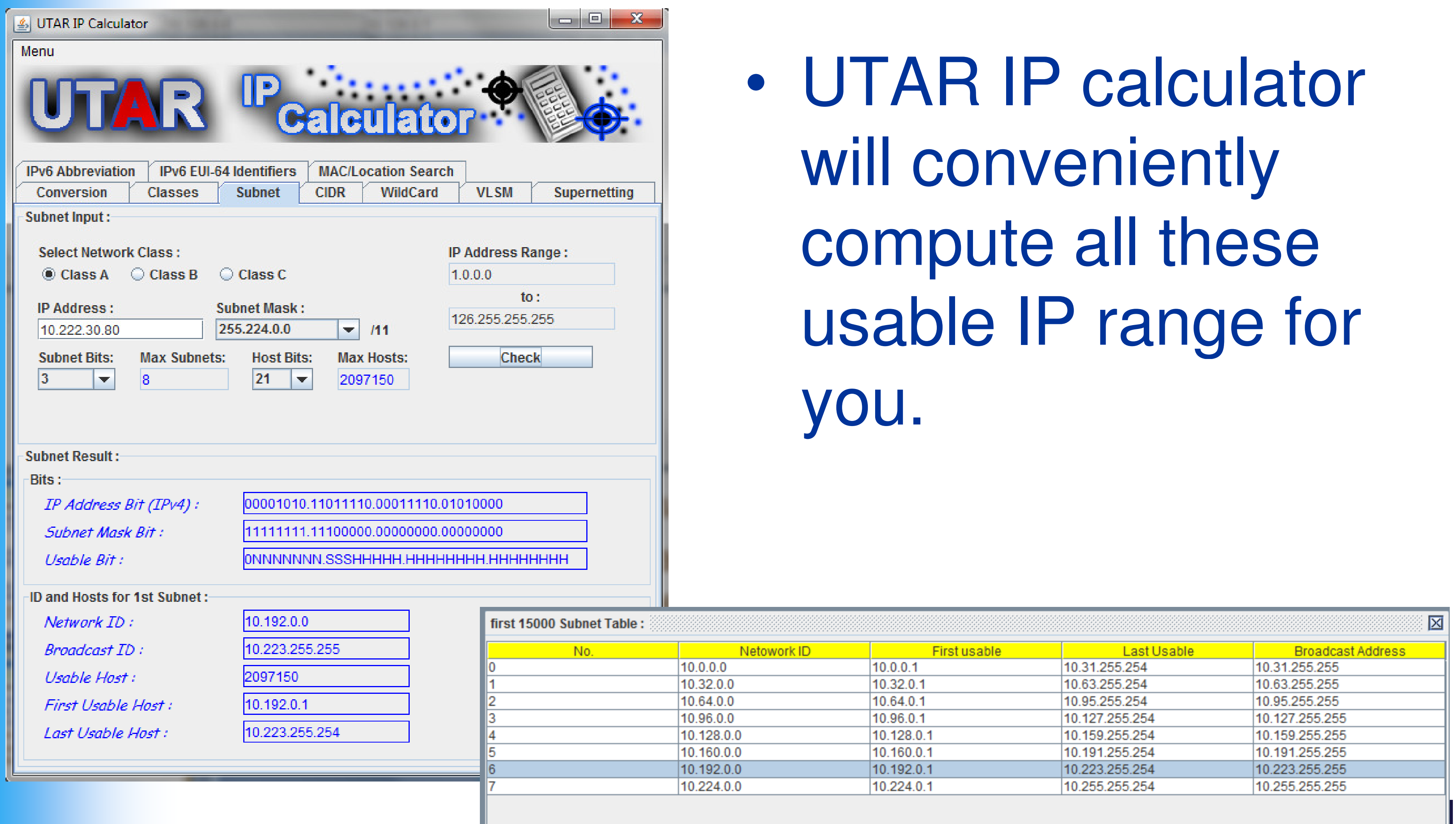
UCCN2243: Usable IP for Host (2)

- Typically, in a subnet, for a full IP address range, at least 3 IP addresses will not be used for hosts, they are:
  - Network address
  - Broadcast address
  - Default gateway IP
- Hence the subnet with mask of /30 is only for point to point network.

UCCN2243: Gateway IP

- In Cisco subnet design,
  - Last usable IP is normally used as the gateway IP
- In Huawei subnet design,
  - First usable IP is normally used as the gateway IP.
- Either way works, since it is just a guideline for easy maintenance.

Demo of UTAR IP Calculator



- UTAR IP calculator will conveniently compute all these usable IP range for you.

IP Subnet Rule #10

UCCN1004: IP Subnet Rules #10

- When Internet addresses were standardized (in early 1980s), the IP addresses were divided up into 5 classes:
- Class A:
  – Network prefix is 8 bits long.
  – Default mask: 255.0.0.0. or /8
- Class B:
  – Network prefix is 16 bits long.
  – Default mask: 255.255.0.0 or /16
- Class C:
  – Network prefix is 24 bits long.
  – Default mask: 255.255.255.0 or /24
- Class D:
  – is multicast address
- Class E:
  – Experimental

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

	First byte	Second byte	Third byte	Fourth byte
Class A	0 to 127			
Class B	128 to 191			
Class C	192 to 223			
Class D	224 to 239			
Class E	240 to 255			

UCCN1004: IP Subnet Rules #10

- We can only use class A, B, and C for host IP address.
  - Class A, B, C IP addresses are called unicast IP address
- We CAN'T use class D and E IP address for "normal" IP address.
  - Class D IP addresses are called multicast IP addresses
  - Class D IP usage is quite different from class A, B, C.

Class	Leading bits	Start	End	CIDR suffix	Default subnet mask
Class A	0	0.0.0.0	127.255.255.255	/8	255.0.0.0
Class B	10	128.0.0.0	191.255.255.255	/16	255.255.0.0
Class C	110	192.0.0.0	223.255.255.255	/24	255.255.255.0
Class D (multicast)	1110	224.0.0.0	239.255.255.255	/4	not defined
Class E (reserved)	1111	240.0.0.0	255.255.255.255	/4	not defined

- *n* indicates a binary slot used for network ID.
- *H* indicates a binary slot used for host ID.
- *X* indicates a binary slot (without specified purpose)

```
Class A
  0.  0.  0.  0 = 00000000.00000000.00000000.00000000
127.255.255.255 = 01111111.11111111.11111111.11111111
                  0nnnnnnn.HHHHHHHH.HHHHHHHH.HHHHHHHH

Class B
128.  0.  0.  0 = 10000000.00000000.00000000.00000000
191.255.255.255 = 10111111.11111111.11111111.11111111
                  10nnnnnn.nnnnnnnn.HHHHHHHH.HHHHHHHH

Class C
192.  0.  0.  0 = 11000000.00000000.00000000.00000000
223.255.255.255 = 11011111.11111111.11111111.11111111
                  110nnnnn.nnnnnnnn.nnnnnnnn.HHHHHHHH

Class D
224.  0.  0.  0 = 11100000.00000000.00000000.00000000
239.255.255.255 = 11101111.11111111.11111111.11111111
                  1110XXXX.XXXXXXXX.XXXXXXXX.XXXXXXXX

Class E
240.  0.  0.  0 = 11110000.00000000.00000000.00000000
255.255.255.255 = 11111111.11111111.11111111.11111111
                  1111XXXX.XXXXXXXX.XXXXXXXX.XXXXXXXX
```

UCCN2243: Classful Issues

- It is an obsolete concept.
- There are three main problems with “classful” addressing,
  - Lack of Internal Address Flexibility:
    - Big organizations are assigned large, “monolithic” blocks of addresses such as class B.
  - Inefficient Use of Address Space:
    - The existence of only three block sizes (classes A, B and C) leads to waste of limited IP address space.
  - Proliferation of Router Table Entries:
    - As the Internet grows, more and more entries are required for routers to handle the routing of IP datagrams, which causes performance problems for routers. Attempting to reduce inefficient address space allocation leads to even more router table entries.

- Classful addressing especially the routing, has evolved into Classless addressing system is also known as CIDR (Classless Inter-Domain Routing).
  - Will be discussed in later lectures.

IP Subnet Rule #11

UCCN1004: IP Subnet Rules #11

- The range of private IP addresses which is NOT used in public IP address for global Internet.
- The Internet Assigned Numbers Authority (IANA) has reserved the following three blocks of the IP address space for private internets (local networks):
  - **10.0.0.0 - 10.255.255.255**
  - **172.16.0.0 - 172.31.255.255**
  - **192.168.0.0 - 192.168.255.255**
- IANA has reserved private the following IP addresses for Automatic Private IP Addressing (APIPA) for Windows platform (except NT).
  - **169.254.0.0 - 169.254.255.255**
- APIPA is used in Windows where the IP address (169.254.x.x) is assigned automatically by the OS (after sometime) when the DHCP service is not available (and the option of “Obtain an IP address automatically” is on).
  - If the DHCP service is working fine in your LAN, and you still get the IP 169.254.x.x for your Windows IP, that only means your cable, connection or NIC is having problem.

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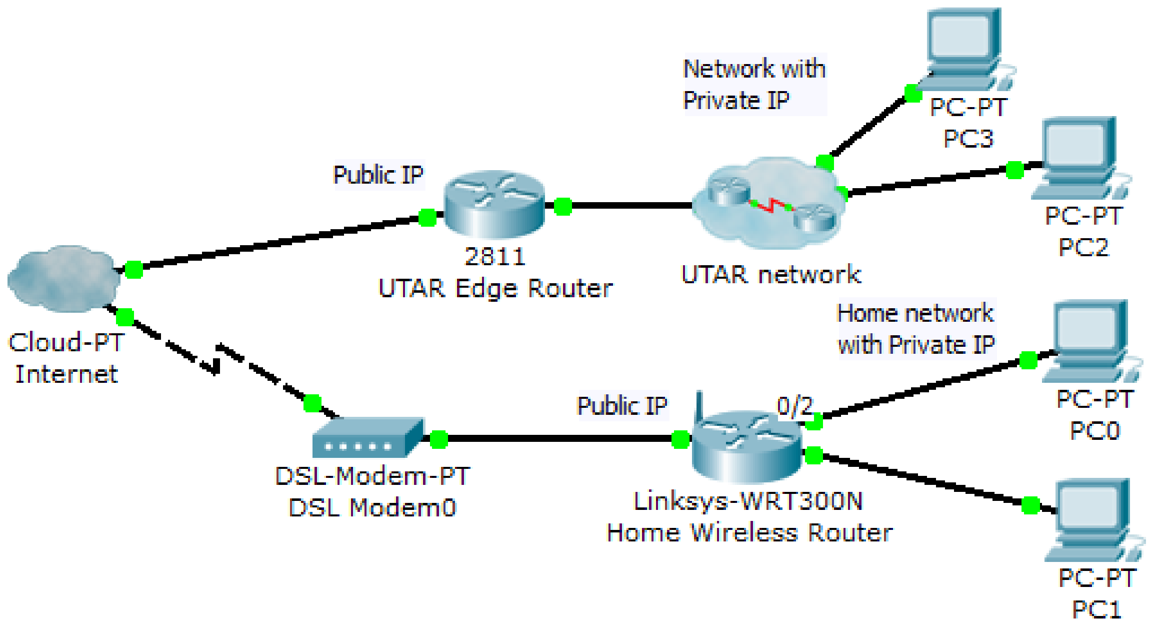
UCCN2003: Reserved, Loopback & Private IP Addresses

Range Start Address	Range End Address	"Classful" Address Equivalent	Classless Address Equivalent	Description
0.0.0.0	**0.255.255.255**	Class A network 0.x.x.x	0/8	Reserved.
10.0.0.0	**10.255.255.255**	Class A network 10.x.x.x	10/8	Class A private address block.
127.0.0.0	**127.255.255.255**	Class A network 127.x.x.x	127/8	Loopback address block.
128.0.0.0	**128.0.255.255**	Class B network 128.0.x.x	128.0/16	Reserved.
169.254.0.0	**169.254.255.255**	Class B network 169.254.x.x	169.254/16	Class B private address block reserved for automatic private address allocation. See the section on DHCP for details.
172.16.0.0	**172.31.255.255**	16 contiguous Class B networks from 172.16.x.x through 172.31.x.x	172.16/12	Class B private address blocks.
191.255.0.0	**191.255.255.255**	Class B network 191.255.x.x	191.255/16	Reserved.
192.0.0.0	**192.0.0.255**	Class C network 192.0.0.x	192.0.0/24	Reserved.
192.168.0.0	**192.168.255.255**	256 contiguous Class C networks from 192.168.0.x through 192.168.255.x	192.168/16	Class C private address blocks.
223.255.255.0	**223.255.255.255**	Class C network 223.255.255.x	223.255.255/24	Reserved.

- The range 127.0.0.0 to 127.255.255.255.
  - IP datagrams sent by a host to a 127.x.x.x loopback address are not passed down to the data link layer for transmission.

UCCN2243: Private & Public IP Address

- Private IP addresses are the IP addresses that use at home (and at school), as your source IP.
- Public IP is global, you need a public IP to go to Internet
- Router (and wireless router) will undergo a process called "network address translation" for a private IP to go to Internet with a public IP.



IP Subnet Rule #12

UCCN1004: IP Subnet Rule #12

- A host can have different IP addresses according to the number of network interface cards installed.
- IP is a Network Interface address
  - A PC with 1 NICs requires 1 IP address
  - A PC with 3 NICs requires 3 IP addresses
    - One IP for each NIC
  - A router with 2 Fast Ethernet ports and 2 T1 serial ports need 4 IP address
    - One IP for each ports
- A DNS server can have 2 NICs with 2 different IP addresses
- From now on, “adjust” your previous thought on IP address is a network “host” address.
  - Meaning 1 IP = 1 host.

IP Subnet Rule #13

UCCN1004: IP Rule #13

- Host ID bits as LAN design parameter for allocating the number of PCs/hosts in a LAN.
  - Similar to enhanced Rule #2 and Rule #4

- For example: For a subnet with a mask of 255.255.255.128, how many hosts that we can allocate in that subnet?
  - 255.255.255.128 => /25 => 32-25 = 7 host bits
  - 2^7 => 128 host ID => theoretically 128 IP address.
  - Actual allocation of PC/host IP = 128 – 1 – 1 – 1 = 125 IP address
  - Can't use network address, broadcast address, and gateway address for PC/hosts. (Assuming 1 gateway in the LAN)
  - Host includes laptops, servers and printers.

- If a LAN is desired to have 27 PCs, what should be the subnet mask?
  - Formula: $2^H \geq$ "number of hosts" + 3; H = host ID bits
  - 3 = 1 network address + 1 broadcast address + 1 gateway
  - $2^H \geq 27 + 3$ => $2^5 \geq 30$ =>
  - H = 5; Subnet mask = /(32-5) = /27 => 255.255.255.224

UCCN2243: IP Subnet Rule #2

- Subnet mask as a design parameters for the number of host in a subnet.

- What is the maximum number of usable address for this subnet with a subnet mask of
  - 255.255.240.0 ?

Answer

- 255.255.240.0
  - There are 8+8+4+0 bits = 20, for network ID
  - There left 32 – 20 = 12 bits for host ID
  - Subnet can fit maximum 2^{12} = 4096
  - Need to minus off the two host ID with all ‘0’ and all ‘1’ in host bits.
  - So answer => 4096 – 2 = 4094.

UCCN2243: Quick Quiz

- If a subnet has 140 hosts,
  - How many network ID bits does this subnet require (in IPv4)?
  - What is the value of the subnet mask?
  - How many usable address is left in this subnet?

Answer

- 140 >= 2^7 (128)
- 140 <= 2^8 (256)
- Answer #1: 32 bits – 8 bits (host) = 24 bits
- Answer #2: 255.255.255.0
- Answer #3: 256 – 140 = 116 hosts
  - A more appropriate answer is 116 – 2 = 114 usable address.
  - 2 is the network address and broadcast address.

IP Subnet Rule #14

Subnetting

UCCN2243: IP Rule #14

- Rule #14 talks about subnetting
  - 1 bigger IP subnet split into a few smaller IP subnets.
  - It involves subnet design and planning.

- It is a long topic to be discussed in full lecture.

- Will proceed to lecture 02…..

IP Subnet Rule #15

Supernetting

UCCN2243: IP Rule #15

- Rule #15 talks about supernetting.
  - 2 or more smaller IP subnets are joined into a larger IP subnet.
- It involves IP address summarization.
- Will be discussed in lecture 02…..

Other Issues of Subnet

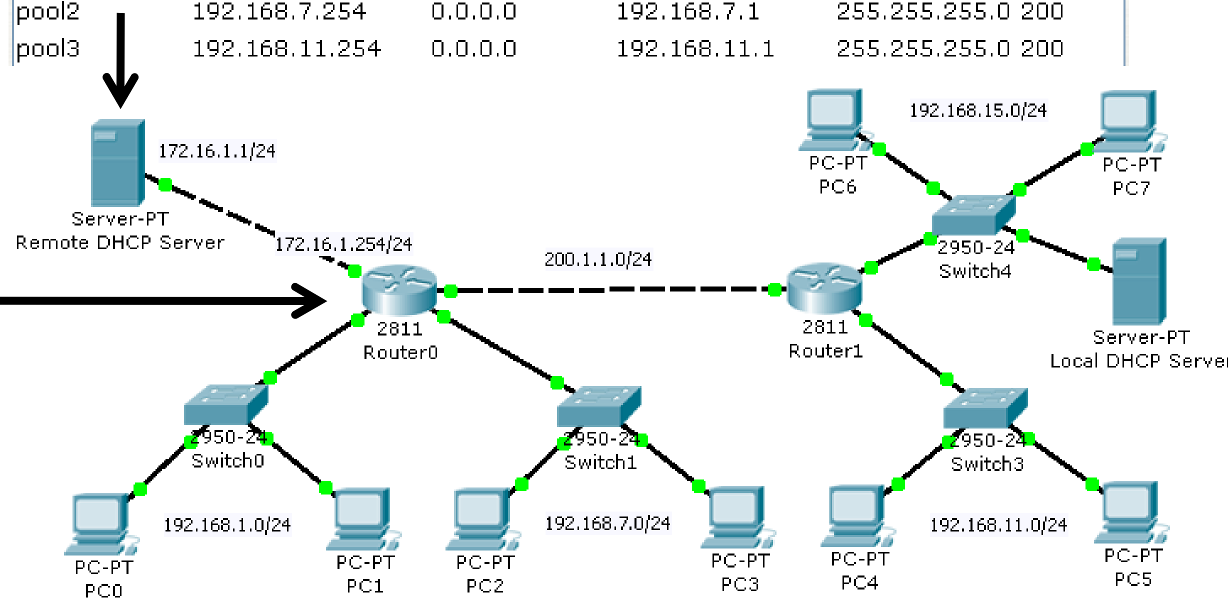
Remote DHCP Service

UCCN2243: Remote DHCP Server

- In UCCN1003, DHCP is in the same LAN
- In UCCN2243, DHCP can be set in a “remote” LAN
  - With the help of the command “ip helper-address *DHCP_server_IP_address”*
  - DHCP Server has to be set with a few and appropriate IP address pool.

```
!
!
interface FastEthernet0/0
 ip address 192.168.1.254 255.255.255.0
 ip helper-address 172.16.1.1
 duplex auto
 speed auto
!
interface FastEthernet0/1
 ip address 192.168.7.254 255.255.255.0
 ip helper-address 172.16.1.1
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 172.16.1.254 255.255.255.0
 duplex auto
 speed auto
!
interface FastEthernet1/1
 ip address 200.1.1.1 255.255.255.0
 duplex auto
 speed auto
!
```

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max Nur
serverPool	0.0.0.0	0.0.0.0	172.16.0.0	255.255.0.0	1408237
pool1	192.168.1.254	0.0.0.0	192.168.1.1	255.255.255.0	200
pool2	192.168.7.254	0.0.0.0	192.168.7.1	255.255.255.0	200
pool3	192.168.11.254	0.0.0.0	192.168.11.1	255.255.255.0	200



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Steps to have remote DHCP service

1. Create multiple pools in the DHCP device (routers or server)
2. Provide routes to the device so that the end devices or gateway router can reach them.
3. Key in the “ip helper-address” in the gateway of the subnet (interface or sub-interface).

UCCN2243: IP helper-address

```
Router(config)#int fa0/0
Router(config-if)#ip helper address 172.16.1.1
```

- Steps to use IP helper-address:
  - Use the command at the interfaces
  - Point the “helper address” to the DHCP server IP address
  - The DHCP server must be configured with the subnet IP range (or DHCP pool).
  - Put the IP helper-address at the router gateway interface.